THE OBSTRUCTION TO A SPLIT INJECTION OF FINITE PROJECTIVE MODULES BEING A DIRECT SUM OF SUBMODULES IS CUT OUT BY A FINITELY GENERATED IDEAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The author closes with 'Some details omitted.'; that sentence is kept verbatim and marks the one step left to the reader.

Allowed standard prerequisites (editorial).

- Finite projective modules and split injections. the hypothesis that s is a split injection at source lines 28129-28130. Supplies the splitting pi used to build the endomorphisms p_i and the module structure that makes the obstruction computable.
- Idempotent and orthogonal projectors on a module. the criterion at proof lines 28141-28145. Turns the geometric condition 'direct sum of submodules' into algebraic equations whose entries generate J.

Lemma 0.2. *Let A be a ring. Let M, N_1, N_2 be finite projective A-modules. Let $s : M \to N_1 \oplus N_2$ be a split injection. There exists a finitely generated ideal $J \subset A$ with the following property: a ring map $A \to B$ factors through A/J if and only if $s \otimes id_B$ identifies $M \otimes_A B$ with a direct sum of submodules of $N_1 \otimes_A B \oplus N_2 \otimes_A B$.*

Proof. Choose a splitting $\pi : N_1 \oplus N_2 \to M$ of s. Denote $q_i : N_1 \oplus N_2 \to N_1 \oplus N_2$ the projector onto N_i. Set $p_i = \pi \circ q_i \circ s$. Observe that $p_1 + p_2 = \mathrm{id}_M$. We claim M is a direct sum of submodules of $N_1 \oplus N_2$ if and only if p_1 and p_2 are orthogonal projectors. Thus J is the smallest ideal of A such that $p_1 \circ p_1 - p_1$, $p_2 \circ p_2 - p_2$, $p_1 \circ p_2$, and $p_2 \circ p_1$ are contained in $J \otimes_A \mathrm{End}_A(M)$. Some details omitted. □

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